Project Initialization and Planning Phase

Date	13 July 2026
Team ID	xxxxxx
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

This proposal report aims to transform credit card approval using machine learning, boosting efficiency and accuracy. It tackles inefficiencies in manual review, promising faster operations, more consistent decisions, and improved applicant experience. Key features include a machine learning-based credit approval model and real-time, form-based predictions.

Project Overview	
Objective	The primary objective is to streamline the credit card approval process by using a machine learning model to assess applicant creditworthiness quickly and consistently, reducing reliance on slow, manual review.
Scope	The project covers data preprocessing and feature engineering on applicant and credit-history data, training and comparing classification models (Logistic Regression, Decision Tree, Random Forest), and deploying the best-performing model behind a Flask web application for real-time approval predictions.
Problem Statement	
Description	Manual review of credit card applications is slow, inconsistent, and prone to human bias, especially as application volume grows, leading to processing delays and inconsistent approval decisions.
Impact	Automating the initial screening step reduces processing time, improves consistency in approval decisions, and frees credit officers to focus on borderline or complex cases, improving both efficiency and applicant experience.
Proposed Solution	
Approach	Train a supervised machine learning classifier on historical application and credit-payment data to predict whether an applicant should be approved, then serve the trained model through a Flask web application with a simple form-based interface.
Key Features	- Machine learning-based credit approval model (Random Forest Classifier, chosen after comparison with Logistic Regression and Decision Tree).

	- Real-time prediction via a simple web form (Flask /predict endpoint), returning an Approved/Rejected result. - Model trained on merged applicant demographic and credit payment-history data, and can be retrained as new data becomes available.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard CPU (no GPU required for training or inference with a Random Forest model)
Memory	RAM specifications	4-8 GB RAM (sufficient for CSV-scale training data)
Storage	Disk space for data, models, and logs	~150 MB (application_record.csv, credit_record.csv, and the serialized model file)
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, joblib, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook
Data		
Data	Source, size, format	Kaggle - Credit Card Approval Prediction dataset: application_record.csv and credit_record.csv (CSV format)